

# Darin Tsui

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## Education

- Georgia Institute of Technology**, Ph.D. in Electrical and Computer Engineering Aug 2023 – May 2028
- Advisor: Amirali Aghazadeh
- University of California San Diego**, B.S. in Bioengineering Sep 2019 – Jun 2023

## Selected Publications

**Darin Tsui**, Aryan Musharaf, Justin S. Kang, Yigit E. Erginbas, and Amirali Aghazadeh. SHAP zero explains biological sequence models with near-zero marginal cost for future queries. In *Advances in Neural Information Processing Systems (NeurIPS)* (Accepted), 2025.

Thomas A. Walton, **Darin Tsui**, Aryan Musharaf, and Amirali Aghazadeh. SpecMER: Fast protein generation with K-mer guided speculative decoding. In *NeurIPS* (Accepted, **Spotlight**), 2025.

**Darin Tsui**, Kunal Talreja, and Amirali Aghazadeh. Sparse autoencoders for low- $N$  protein function prediction and design. In *NeurIPS AI4Science Workshop* (Accepted), 2025.

## Honors & Awards

- NSF Graduate Research Fellowship** 2024 - 2027
- Research proposal title: *Disease Variant Prediction Using Deep Generative Models*.
- Course Enhancements & Improvement (CREATION) Award**, Georgia Tech 2023
- Award received for developing the course ECE 8803—*Generative and Geometric Deep Learning*.

## Experience

- AI ML and Information Research Group**, Georgia Tech – Atlanta, GA Jun 2023 – Present
- Explained models 1000-fold quicker than state-of-the-art methods by designing algorithms at the intersection of *signal processing, coding theory, combinatorics, and interpretability*.
  - Leveraged *mechanistic interpretability* methods, such as sparse autoencoders and cross-layer transcoders, for extracting higher-order interactions, predicting protein function, and designing novel protein variants in the biosciences.
  - Designed and developed principled *generative models* that leverage domain priors.
- Data Engineering Intern**, Surgalign – San Diego, CA Jan 2023 - Aug 2023
- Interned at Surgalign, a small medical device company focused on developing AI solutions in spinal surgery (now acquired by Xtant Medical and Augmetics).
  - Designed an internal MRI image processing application using Python that reduced preprocessing time.
  - Assisted in developing ground truth dataset of imaging data and validated deep learning models against surgeon-generated data using Sørensen-Dice coefficient statistical testing.

## Leadership

- Reviewer**: 2025 NeurIPS AI4Science Workshop, 2024 ICLR GEM Workshop.
- Capstone Project Advisor**, UC San Diego – San Diego, CA Jul 2023 - Jun 2024
- Capstone project advisor for a bioengineering capstone project with Dr. Frank E. Talke.
  - Mentored two undergraduates toward the design and development of a surgical navigation system with 3D tracking and augmented reality.
  - Capstone project team won the *Teaching Assistant's Choice Award* at Bioengineering Day.